

```
import nltk
import matplotlib.pyplot as plt
```

```
s = "hello all how are you all doing"
```

```
from nltk.tokenize import word_tokenize,sent_tokenize
```

```
nltk.download('punkt')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
True
```

```
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
True
```

```
nltk.download('average_perceptron_tagger')
```

```
[nltk_data] Error loading average_perceptron_tagger: Package
[nltk_data]   'average_perceptron_tagger' not found in index
False
```

```
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
True
```

```
nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]   /root/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data]   date!
True
```

```
wt = word_tokenize(s)
```

```
print(wt)
```

```
['hello', 'all', 'how', 'are', 'you', 'all', 'doing']
```

```
st = sent_tokenize(s)
```

```
print(st)
```

```
['hello all how are you all doing']
```

```
from nltk.probability import FreqDist
```

```
fd = FreqDist(s)
```

```
print(fd)
```

```
<FreqDist with 14 samples and 31 outcomes>
```

```
fc = FreqDist(tt.lower() for tt in wt)
```

```
fc
```

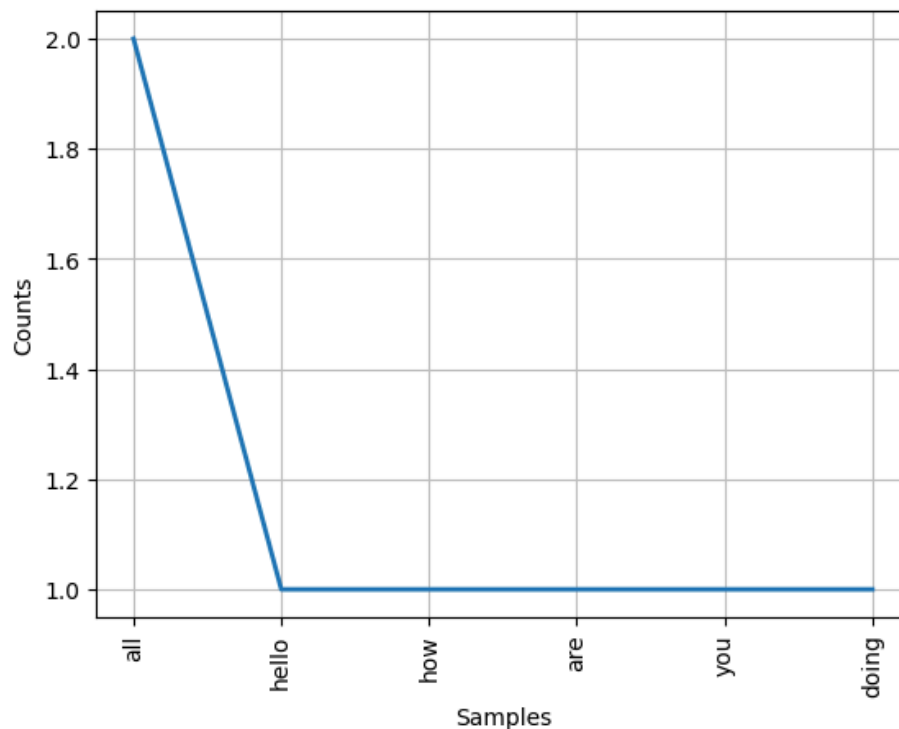
```
FreqDist({'all': 2, 'hello': 1, 'how': 1, 'are': 1, 'you': 1, 'doing': 1})
```

```
fc.most_common(3)
```

```
[('all', 2), ('hello', 1), ('how', 1)]
```

```
fc.plot()
```

```
<Axes: xlabel='Samples', ylabel='Counts'>
```



```
from nltk.corpus import stopwords
```

```
stop_words = set(stopwords.words('english'))
```

```
filtered_tokens = [word for word in wt if word not in stop_words]
```

```
print("Filtered:", filtered_tokens)
```

```
Filtered: ['hello']
```

```
from nltk import pos_tag
```

◆ Gemini

```
nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
```

```
[nltk_data] /root/nltk_data...
```

```
[nltk_data] Package averaged_perceptron_tagger_eng is already up-to-
```

```
[nltk_data]      date!  
True
```

```
pt = pos_tag(wt)
```

```
for word, pos_tag in pt:  
    print(f"{word}:{pos_tag}")
```

```
hello:VB  
all:DT  
how:WRB  
are:VBP  
you:PRP  
all:DT  
doing:VBG
```

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